

Deep Learning

Assignment 1

Perform PCA in Dimension Reduction of Numerical Data

1. Pre-process the data through standardization
2. Perform PCA to reduce dimension
3. Construct the scree plot
4. Data visualization in lower dimensional representation

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Import necessary libraries for data manipulation, plotting, and PCA

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
```

Load the Iris dataset and display the first few rows

```
from sklearn.datasets import load_iris

iris = load_iris()
X = iris.data      # numerical features
y = iris.target    # class labels

df = pd.DataFrame(X, columns=iris.feature_names)
df.head()
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2

Initialize a StandardScaler and apply it to standardize the feature data

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

Perform PCA with 2 components on the scaled data and print the resulting shape

```
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

print("Shape after PCA:", X_pca.shape)
```

Shape after PCA: (150, 2)

Print the explained variance ratio for each principal component

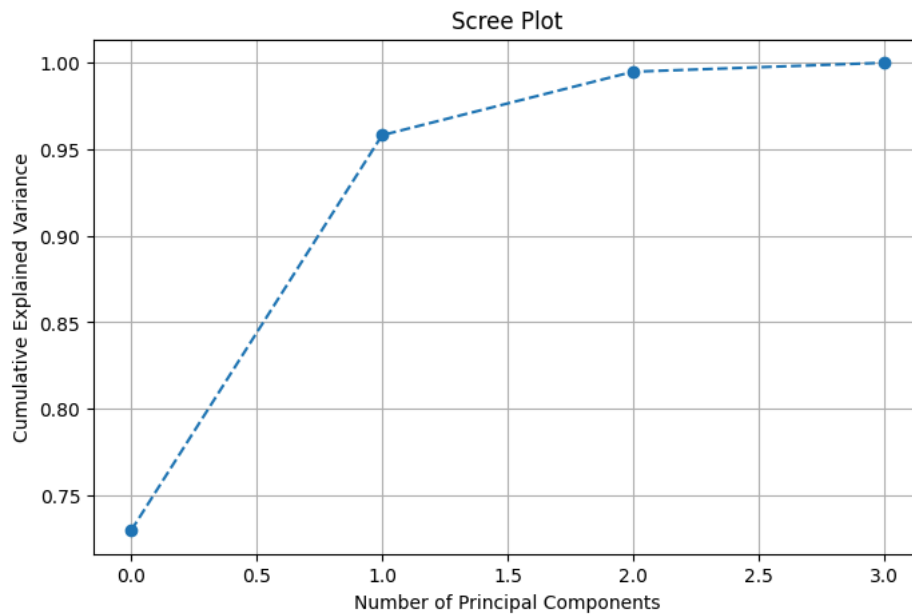
```
print("Explained Variance Ratio:", pca.explained_variance_ratio_)
```

Explained Variance Ratio: [0.72962445 0.22850762]

Perform PCA on the full dataset to construct and display a scree plot

```
pca_full = PCA()
X_pca_full = pca_full.fit_transform(X_scaled)

plt.figure(figsize=(8,5))
plt.plot(
    np.cumsum(pca_full.explained_variance_ratio_),
    marker='o',
    linestyle='--'
)
plt.xlabel("Number of Principal Components")
plt.ylabel("Cumulative Explained Variance")
plt.title("Scree Plot")
plt.grid()
plt.show()
```



Visualize the 2D PCA-transformed data, coloring points by their target class

```
plt.figure(figsize=(8,6))

for target in np.unique(y):
    plt.scatter(
        X_pca[y == target, 0],
        X_pca[y == target, 1],
        label=iris.target_names[target]
    )

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("PCA - 2D Data Visualization")
plt.legend()
plt.grid()
plt.show()
```

